The MCI Takeover Battle

Verizon versus Qwest

A plain-English study guide — everything from scratch, with worked numbers and final results

Financial Analysis of Corporate Deals • HBS 9-206-045

The one puzzle this whole case is about:

MCI's board had two offers. Qwest's looked bigger on paper ($26 a share vs Verizon's $23.50). The board chose Verizon — the *lower* offer. Everything below builds, step by step, to why that was the right call.

What's inside

Read it top to bottom the first time — each part uses the one before it.

- **Part 1 —** Why getting paid in shares is risky (the whole foundation)
- **Part 2 —** The four payoff shapes: simple swap, floor, cap, collar
- **Part 3 —** Reading the two real offers (Verizon = a floor, Qwest = a collar)
- **Part 4 —** Putting a dollar value on each offer
- **Part 5 —** The option formula (Black-Scholes) with the z-table
- **Part 6 —** Why the board chose Verizon (the answer to the puzzle)
- **Part 7 —** All the final numbers in one place
- **Part 8 —** Exam watch-outs (the traps)

The story in one page

MCI is a telecom company (born 1963 as Microwave Communications). WorldCom bought it in 1998, WorldCom went bankrupt in 2002 after an accounting fraud, and MCI came back out of bankruptcy in 2004 — now with very little debt and about $5.5 billion of cash. Two rivals wanted to buy it.

Verizon — a giant, financially rock-solid phone company (top-quality A+ credit rating, market value about $96.6 billion, steadily profitable). Its shares barely move (19% volatility) and it pays a dividend.

Qwest — a much smaller, weaker player with a fraud history, about $17.3 billion of debt, negative net worth (it owed more than it owned), and a shaky B+ credit rating. Its shares are very jumpy (45% volatility) and it pays no dividend.

The bidding (Feb–Mar 2005): Qwest and Verizon both bid; MCI's board first sided with Verizon even though Qwest offered more. Qwest then raised its bid to $26 a share; on 29 March 2005 Verizon revised its offer to $23.50 a share, and the board had to decide again.

The numbers you'll keep using (as of 29 March 2005):

Item	Value
MCI shares outstanding	≈ 326 million
MCI share price	$23.78
Verizon share price	$34.86 (volatility 19%, dividend yield 4.4%)
Qwest share price	$3.79 (volatility 45%, no dividend)
Risk-free rate (1-year)	≈ 3.4%
Time until the deal closes	≈ 1 year
MCI corporate tax rate	35%

Part 1 — Why getting paid in shares is risky

When a buyer pays you in cash, you know exactly what you're getting the moment you sign. When a buyer pays you in their own shares, you don't — because the deal takes about a year to actually close, and over that year the buyer's share price can move a lot. If the number of shares is fixed, the value you

finally receive floats up and down with the buyer's stock. That single fact is what the whole case is about.

A corner-shop version. *You're selling your shop. The buyer pays you in their company's shares and the deal closes in a year. You've agreed the payment should be worth ₹100. Their share price today is ₹10, so the plan is to hand you 10 shares (10 × ₹10 = ₹100). But if their price drops to ₹6 by closing, your 10 shares are worth only ₹60. If it rises to ₹15, they're worth ₹150. You're along for the ride — up and down.*

To manage that ride, buyers and sellers add protection features. A floor protects the seller, a cap protects the buyer, and a collar splits the risk between them. Those three features are just the four shapes in Part 2.

Part 2 — The four payoff shapes

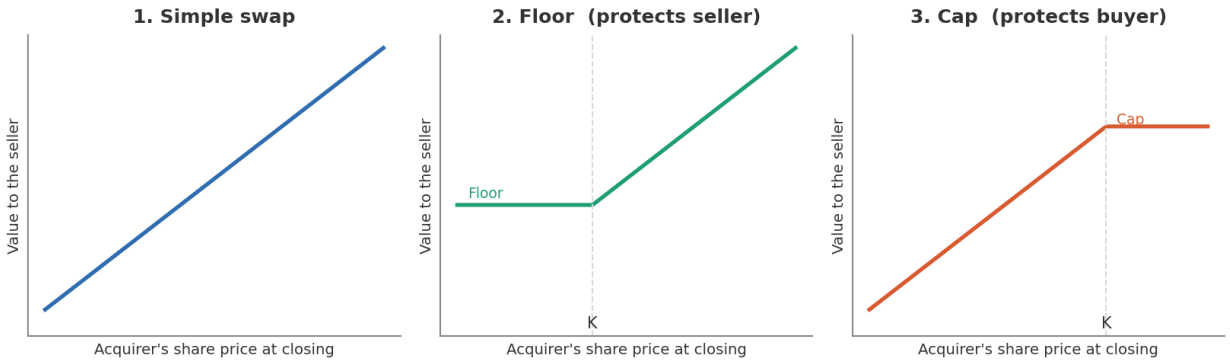
In every picture below, the bottom axis is the buyer's share price at closing, and the up axis is the value the seller ends up receiving.

Simple swap, floor, and cap

Simple swap: a fixed number of shares. Value just rises in a straight line with the buyer's price. No protection — pure ride.

Floor (protects the seller): "you'll never get less than a set minimum." If the price falls, the seller is handed extra shares so the value can't drop below the floor — but the seller keeps all the upside if the price rises. Its shape is flat, then rising.

Cap (protects the buyer): "the seller will never get more than a set maximum." If the price rises, the seller is handed fewer shares so the value can't climb past the cap. Its shape is rising, then flat.



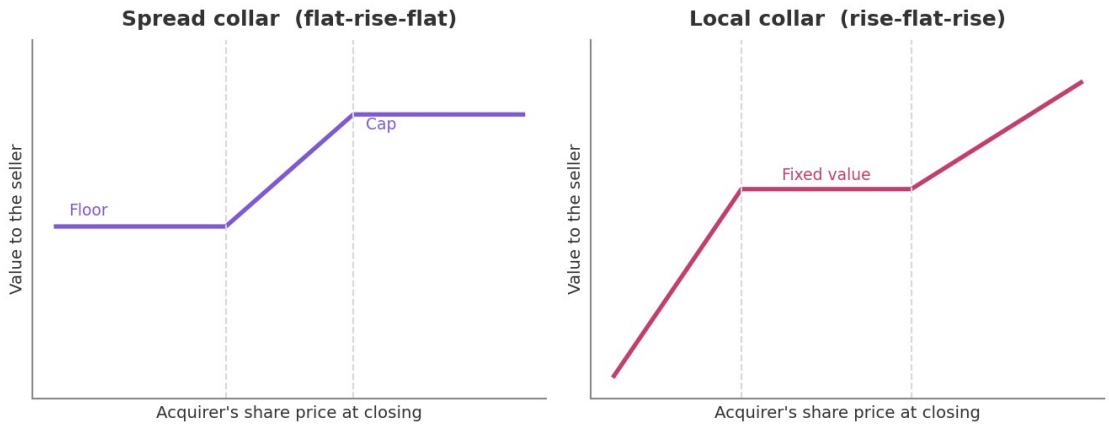
The three building blocks. 'K' marks the kink — the price where protection switches on or off.

Collars (a floor and a cap together)

A collar gives the seller downside protection but makes them give up some upside — a floor and a cap at once. There are two versions, and telling them apart is a classic exam question.

Spread collar: flat at both ends, rising in the middle. The cash value is fixed at the extremes, and the number of shares is fixed in the middle band.

Local collar: the opposite — flat in the middle, rising at both ends. The value is pinned to a fixed amount across a middle band, and floats outside it (up above the band, down below it). This is Qwest's structure.



Memory hook: spread collar is flat at the ENDS; local collar is flat in the MIDDLE.

> **Who benefits from what — never get this backwards:**
> Floor → the seller (target). Cap → the buyer (acquirer). Collar → both.

Part 3 — Reading the two real offers

Now we map the actual offers onto those shapes. Two useful rules for reading the fine print:

- **"Adjusted to deliver a value of $X"** → the value is fixed and the share count floats → that stretch is FLAT.
- **"The exchange ratio shall be [a number]"** → the share count is fixed and the value floats → that stretch SLOPES.

Verizon = a floor

Verizon pays $8.75 in cash plus "$14.75 worth of Verizon shares," and promises "never fewer than 0.4062 shares." The number of shares you get is normally $14.75 ÷ (Verizon's price), but it can't drop below 0.4062. Watch what that does:

Verizon price at closing	Shares (14.75 ÷ price, min 0.4062)	Stock value	Total (+ $8.75 cash)
$20	0.7375	$14.75	$23.50
$30	0.4917	$14.75	$23.50
$36.31	0.4062	$14.75	$23.50
$40	0.4062 (floor kicks in)	$16.25	$25.00
$50	0.4062 (floor kicks in)	$20.31	$29.06

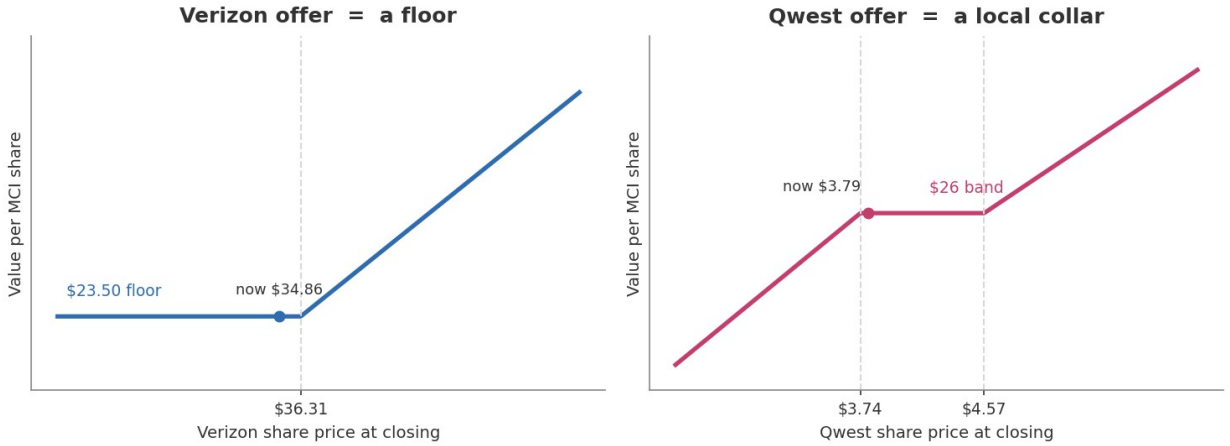
So at $36.31 and below you always get exactly $23.50 (your safety net); above $36.31 you ride the upside. Where does $36.31 come from? It's simply the price at which 0.4062 shares equals $14.75 (0.4062 × 36.31 = $14.75). In one line: **you're guaranteed at least $23.50, and you get more if Verizon rises above $36.31.**

Qwest = a local collar

Qwest pays $10.50 cash plus "$15.50 worth of Qwest shares." The share count floats to keep you at $15.50 as long as Qwest stays between $3.74 and $4.57. Below $3.74 the count freezes (4.1444 shares) so your value falls; above $4.57 it freezes (3.392 shares) so your value rises.

Qwest price at closing	Shares you get	Stock value	Total (+ $10.50 cash)
$3.00	4.1444 (frozen)	$12.43	$22.93
$3.50	4.1444 (frozen)	$14.51	$25.01
$3.74	4.1444	$15.50	$26.00
$4.15	3.735	$15.50	$26.00
$4.57	3.392	$15.50	$26.00
$5.00	3.392 (frozen)	$16.96	$27.46
$6.00	3.392 (frozen)	$20.35	$30.35

Three zones: below $3.74 you lose (protection is off), between $3.74 and $4.57 you get exactly $26, above $4.57 you gain. Here are both offers drawn with their real numbers:



Verizon protects you all the way down. Qwest only protects you inside a narrow band — and the bottom edge ($3.74) is sitting right under today's price ($3.79).

> **The key difference (remember this for the essay):**
> Verizon's floor protects you no matter how far it falls. Qwest's collar only protects you inside a tight band, and Qwest trades right on the bottom edge — so its "$26" is fragile in a way Verizon's "$23.50" is not.

Part 4 — Putting a dollar value on each offer

The recipe: value each piece separately, then add. A guaranteed amount is shrunk ("discounted") back to today because money in a year is worth a bit less now. A "maybe" (an upside or a downside) is valued with the option formula in Part 5 — for now, just take those option prices as given.

Note on the discount rate. *The professor's answer key discounts the guaranteed cash at the 1-year Treasury rate (about 3.4%), giving $22.73 and $25.14 below. In the walk-through we rounded to 3.35% (which gives ~$25.16 for Qwest) — the difference is a penny or two. Use the key's rate in the exam.*

Valuing Verizon (the easy one — floor only)

Piece	What it is	Value today
Guaranteed $23.50	cash + floor, discounted ~1 year	$22.73
Bonus: 0.4062 × $1.82	the upside above $36.31	$0.74
Value per MCI share		**$23.47**
× ~326 million shares		**≈ $7.66 billion**

Notice it lands a touch below the $23.50 headline — that's the one-year wait, and the thin bonus doesn't quite close the gap.

Valuing Qwest (three pieces — the new move is subtracting the downside)

From the seller's point of view, the upside is a good "maybe" (add it) and the downside is a bad "maybe" (subtract it).

Piece	What it is	Value today
$26 baseline	discounted ~1 year	$25.14
Upside: 3.392 × $0.46	the call, above $4.57	+ $1.56
Downside: 4.1444 × $0.58	**the short put, below $3.74**	**− $2.40**
Value per MCI share		**≈ $24.31**
× ~326 million shares		≈ $7.93 billion

The downside ($2.40) is bigger than the upside ($1.56) — that is Qwest's risk showing up in the arithmetic. It sits right on its $3.74 trapdoor, so falling out the bottom is the likelier, more valuable move. That drags the offer's value down by about $0.84, so the real $24.31 sits far below the $26 headline.

> **The surprise:**
> On the headline, Qwest ($26) beats Verizon ($23.50) by $2.50. After valuing the fine print and discounting, that lead shrinks to just $0.84 ($24.31 vs $23.47) — because Qwest's options subtract value while Verizon's add value. Qwest is still higher on pure money — yet the board picked Verizon. Part 6 explains why.

Part 5 — The option formula (Black-Scholes)

This is the one formula in the case. It prices a "maybe" — an upside (a call) or a downside (a put). It's five inputs, two lookups in the z-table, and your calculator doing the rest.

What it's doing, in plain words: a call pays only sometimes — when the stock ends high enough. So its value is roughly how much you'd collect, weighted by the chance you collect it, minus the discounted strike you'd pay, weighted by the chance you pay it. The N(...) terms are those "chance" weights.

The five inputs (all from the case)

- **S =** the buyer's share price today (Verizon $34.86; Qwest $3.79).
- **K =** the strike — the price where the bonus kicks in ($36.31; $4.57; $3.74).
- **T =** time to closing ≈ 1 year.
- **r =** the risk-free rate ≈ 3.35–3.4%.
- **σ (sigma) =** volatility — how jumpy the stock is (Verizon 19%; Qwest 45%).
- **q =** dividend yield — only if the stock pays dividends (Verizon 4.4%; Qwest 0).

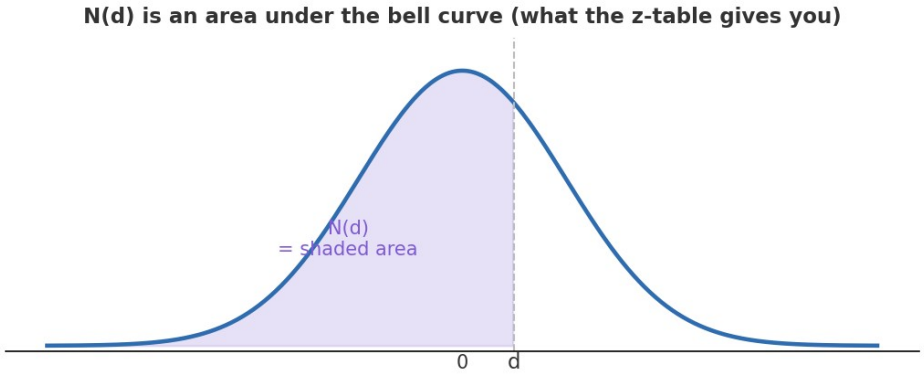
The formula (for a call)

$$d1 = [\ \ln(S \div K) + (r - q + \sigma^2/2) \times T\ ] \div (\sigma \times \sqrt{T})$$

$$d2 = d1 - \sigma \times \sqrt{T}$$

$$Call = S \times e^{(-qT)} \times N(d1) - K \times e^{(-rT)} \times N(d2)$$

N(d) is the one thing the z-table gives you: the area under the bell curve to the left of d — the chance of landing below d.



Reading the z-table means finding this shaded area.

Reading the z-table: it gives N(z) for z ≥ 0 (row = first decimal, column = second). For a negative value use the mirror rule N(−z) = 1 − N(z). Example: N(0.57) = 0.7157, so N(−0.57) = 1 − 0.7157 = 0.2843.

Worked example: the Qwest call (clean — no dividend)

Inputs: S = 3.79, K = 4.57, σ = 0.45, r = 0.0335, T = 1, q = 0.

Step	Value
S ÷ K = 3.79 ÷ 4.57	0.8293

Step	Value
ln(0.8293)	−0.1872
$\sigma^2/2 = 0.45^2 \div 2$	0.10125
$(r + \sigma^2/2)\cdot T = 0.0335 + 0.10125$	0.13475
Top of d1 = −0.1872 + 0.13475	−0.05245
$\sigma\cdot\sqrt{T} = 0.45 \times 1$	0.45
d1 = −0.05245 ÷ 0.45	−0.1166
d2 = −0.1166 − 0.45	−0.5666
N(d1) = 1 − N(0.1166)	0.4536
N(d2) = 1 − N(0.5666)	0.2855
e^(−rT) = e^(−0.0335)	0.9671
Call = 3.79×0.4536 − 4.57×0.9671×0.2855	**1.719 − 1.262 = $0.46**

The Verizon call — same steps, plus the dividend

Verizon pays a dividend, and you're promised the shares a year out, so you miss a year of dividends. That lost income shrinks the option — it's the −q inside d1 and the e^(−qT) factor. Skip it and you'd wrongly get about $2.52 instead of $1.82.

Step	Value
ln(34.86 ÷ 36.31)	−0.0408
$(r - q + \sigma^2/2)\cdot T = 0.0335 - 0.044 + 0.01805$	0.00755
$\sigma\cdot\sqrt{T}$	0.19
d1 = (−0.0408 + 0.00755) ÷ 0.19	−0.175
d2 = −0.175 − 0.19	−0.365
N(d1) = 1 − N(0.175)	0.4305
N(d2) = 1 − N(0.365)	0.3575
e^(−qT) = e^(−0.044)	0.9570
e^(−rT) = e^(−0.0335)	0.9671
Call = 34.86×0.9570×0.4305 − 36.31×0.9671×0.3575	**14.36 − 12.55 ≈ $1.82**

The Qwest put — same steps, but a put pays when the stock ends LOW

So it uses the mirror areas N(−d1) and N(−d2). Inputs: S = 3.79, K = 3.74, σ = 0.45, r = 0.0335, q = 0.

Step	Value
ln(3.79 ÷ 3.74)	+0.0133
$(r + \sigma^2/2)\cdot T$	0.13475
d1 = (0.0133 + 0.13475) ÷ 0.45	+0.329
d2 = 0.329 − 0.45	−0.121
N(−d1) = 1 − N(0.329)	0.3711

Step	Value
N(−d2) = N(0.121)	0.5482
Put = 3.74×0.9671×0.5482 − 3.79×0.3711	**1.983 − 1.407 = $0.58**

The put ($0.58) beats the call ($0.46) because Qwest sits right on its floor — the "bad maybe" is the likelier one.

Part 6 — Why the board chose Verizon

On pure dollars, Qwest ($24.31) beat Verizon ($23.47). Yet the board picked Verizon. Here is the reasoning — and why it holds up.

The board's duty (Revlon)

When a company is being sold, its board must act in shareholders' best interests. In an all-cash deal that usually means simply taking the highest price. But in a stock deal the board has more room: it can weigh the QUALITY of what's being paid, not just the number — because you're being paid in the buyer's shares, so what those "dollars" are really worth depends on how solid the buyer is.

Quality of consideration — the heart of it

Verizon: financially strong (A+ rated, about $96.6 billion market value, steadily profitable), and its shares barely move (19% volatility). MCI is tiny next to Verizon — "a drop in the ocean" — so the Verizon stock MCI's holders receive is stable and dependable.

Qwest: financially weak — negative net worth (about −$2.6 billion), roughly $17.3 billion of debt, a fraud history, and net losses. Its shares are more than twice as jumpy (45% volatility). MCI's shareholders would end up owning a big slice of a fragile, heavily indebted company, and Qwest's price protection only holds inside a narrow band sitting right at today's price — one slip and MCI eats the downside.

Two more points worth a sentence

- **Who was pushing Qwest:** hedge funds and arbitrageurs (large holders like Carlos Slim at 13% and Samuel Heyman at 9%) wanting a quick profit from the higher headline number.
- **"Cash is king":** the market may not have fully priced in Qwest's weakness, so its shares — and the value of its offer — could disappoint.

> **The bottom line (memorise this):**
> A smaller amount of solid, reliable value can be worth more than a larger amount of shaky value. The board judged Verizon's $23.47 of dependable value better than Qwest's $24.31 of fragile value — a defensible decision under its duty to shareholders.

Part 7 — All the final numbers in one place

The two offers, valued

	Verizon	Qwest
Structure	Floor (call only)	Local collar (call + short put)
Headline offer	$23.50	$26.00
Guaranteed / baseline, discounted	$22.73	$25.14
Upside (call)	+ 0.4062 × $1.82 = $0.74	+ 3.392 × $0.46 = $1.56
Downside (short put)	none	− 4.1444 × $0.58 = $2.40
Value per MCI share	**$23.47**	**$24.31**
Total deal value	≈ $7.66 billion	≈ $7.93 billion

	Verizon	Qwest
Board's choice	CHOSEN	Rejected

The three option values

Option	Inputs (S, K, σ, dividend, 1 year)	Value
Verizon call	$34.86, $36.31, 19%, 4.4%	$1.82
Qwest call	$3.79, $4.57, 45%, none	$0.46
Qwest put	$3.79, $3.74, 45%, none	$0.58

Final answer to the puzzle: Qwest's offer was worth more on paper and even after valuation ($24.31 vs $23.47), but the MCI board chose Verizon because Verizon's payment — its stock — was far more solid and reliable than Qwest's. Quality beat quantity.

Part 8 — Exam watch-outs

- **Don't discount an option twice.** Black-Scholes already gives today's value.
- **Include the dividend for Verizon (q = 4.4%).** Skip it and the call comes out ~$2.52 instead of ~$1.82.
- **Upside = add (call); downside = subtract (short put).** Adding the downside by mistake gives a silly ~$29 for Qwest.
- **"Deliver value of $X" = flat; "ratio shall be [number]" = sloped.** That one distinction decides the shape.
- **Strike = protected value ÷ fixed ratio; number of options = the fixed ratio.** This is where $36.31, $4.57 and $3.74 come from.
- **z-table: N(−z) = 1 − N(z); d2 = d1 − σ√T (minus); a put uses N(−d1), N(−d2).** Carry 4 decimals, round only at the end.
- **A board can pick the lower bid in a stock deal.** Quality of consideration / Revlon — this is the whole point of the case.
- **Headline value is always bigger than present value.** Discounting plus the collar's downside explain the gap.

End of study guide — good luck on Monday.